Explainable Federated Stacking Models with Encrypted Gradients for Secure Kidney Medical Imaging Diagnosis

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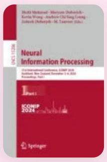

Neural Information Processing

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Abstract

In the field of medical data analysis, both privacy retention and model interpretability are of utmost importance. This paper focuses on the vulnerability of kidney medical image analysis through Federated Learning (FL) with privacy-preserving measures and explainable artificial intelligence (XAI). We introduce a new set of proposals termed Federated Stacking Fusion with Encryption (FedStackEncFL), which integrates the predictive capabilities of ResNet-101 and InceptionV3 architectures through a stacking fusion technique. This approach also guarantees the quality and reliability of the features extracted while preserving the federated data's privacy by encrypting it through Cheon-Kim-Kim-Song (CKKS) during Federated Averaging (FedAvg). Moreover, to optimize privacy, during the computation, the method of adding Gaussian gradient noise is used. The efficiency of the developed technique is proven by the complex experimental data proving the efficiency of the humanitarian approach to enhance indicators of model precision, recall, F1-score, and accuracy in Kidney disease diagnosis. Furthermore, the integration of XAI techniques like GradCAM, GradCAM++, and ScoreCAM provides insightful visual explanations, enhancing the interpretability of our model's predictions in medical diagnostics.

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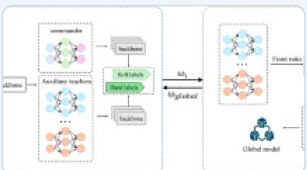

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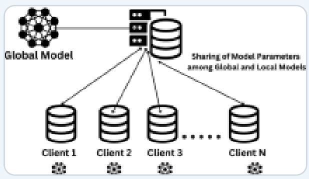

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